

SAT Math Mastery · Week 1A

Linear Equations: Structure, Rearranging & Translating

Time 120 minutes: 90 min solving + 30 min self-check and error log

Topics solution count (one / none / infinite) · structure before solving · solving for an expression · rearranging formulas · translating words · identities · linear inequalities in one and two variables

Name / Date _____

1 · SKILLS & IDEAS

- S1.** Solve any linear equation (distribute, clear fractions with the LCD, collect, divide).
S2. Look at the *structure* first: spot a repeated block and solve for the block or for a target expression.
S3. Decide *one / none / infinitely many* solutions by comparing coefficients, including with parameters.
S4. Match coefficients when two expressions are equal for every x .
S5. Rearrange formulas (make any letter the subject, even when it appears twice).
S6. Translate words: “less than”, “at least”, consecutive integers and multiples, fixed fee plus rate, ages, mixtures.
S7. Linear inequalities: sign reversal, compound inequalities, integer counts, rounding to a feasible whole number.
S8. Two-variable inequalities: test points, read a shaded half-plane or region, count lattice points.

2 · REQUIRED FACTS

- Equal operations on both sides keep an equation true; multiplying or dividing an inequality by a negative number reverses it.
- “ n less than m ” means $m - n$; “at least” is $\geq$; “at most” and “no more than” are $\leq$.
- Consecutive integers differ by 1; consecutive even or odd integers differ by 2; consecutive multiples of 3 differ by 3.
- Every linear equation reduces to $mx = n$: one solution if $m \neq 0$, none if $m = 0$, $n \neq 0$, infinitely many if $m = n = 0$.
- In a graph of an inequality, a dashed boundary means $<$ or $>$ and a solid boundary means $\leq$ or $\geq$; the shading is the side that makes the inequality true.
- A count of whole objects must be a whole number: round *up* for “at least”, *down* for “at most”.

3 · REQUIRED FORMULAS

$$ax + b = cx + d \implies (a - c)x = d - b, \quad x = \frac{d - b}{a - c} \quad (a \neq c)$$

$$\text{total} = \text{fixed fee} + \text{rate} \times \text{quantity}, \quad \text{mixture: amount of solute} = \text{concentration} \times \text{volume}$$

$$n, n + 2, n + 4 \quad (\text{consecutive odd or even}), \quad \text{sum of } k \text{ terms} = k \cdot (\text{mean}), \quad ax + by \leq c \quad (\text{half-plane})$$

4 · MINI THEORY SHEET

Master reduction. Every linear equation collapses to $mx = n$:

$m \neq 0$	one solution	$x = \frac{n}{m}$
$m = 0, n \neq 0$	no solution	$0 = n$ is false
$m = 0, n = 0$	infinitely many	$0 = 0$ is true for every x

Structure before solving. Look for a block that repeats or a factor you can pull out. If $cE = k$ then $E = \frac{k}{c}$; you never need x itself when the question asks for E .

Identity in x . If two expressions are equal for every x , the x -coefficients match *and* the constants match. A constant that multiplies another constant (cd) is still just a constant.

Rearranging. Collect every term containing the target letter on one side, factor it out, then divide. Dividing by a bracket that contains letters is allowed when it is nonzero.

Inequalities. Same steps as equations; reverse only on multiplying or dividing by a negative. For a compound inequality, do the same operation to all three parts. A solution set written “ $x \leq c$ ” tells you the sign of the coefficient you divided by.

Contexts. Fixed + rate: $y = \text{fee} + \text{rate} \cdot n$. For “at most” counts solve then round down; for “at least” round up, and always check the boundary values.

Two-variable inequalities. Draw the boundary line, dashed for strict, solid otherwise. Test one point not on the line; a system is the overlap of the shaded regions.

Speed shortcuts.

Situation	Shortcut
asked for $x + y$ or $2x - y$	add or subtract the equations; never solve for x and y
asked for a block such as $2x - 5$	factor the given expression to show the block
none / infinite with a parameter	set the x -coefficients equal, then compare the constants
true for all x	plug in $x = 0$ and $x = 1$ to get two equations
“at least k less than E ”	$\geq E - k$ (the k is subtracted from E)
solution set given, find a constant	the sign of the divisor decides the direction of the inequality
“could NOT be” with ranges	find the open or closed interval of values; endpoints of strict ranges are excluded
which region / which inequality	test one point per boundary; dashed means strict
whole-number answer	solve, round the correct way, then verify both neighbours

5 · GUIDED EXAMPLES

Example 1 — Structure first: solution count with a parameter

For what value of a does $3(ax - 2) = 12x - 6$ have infinitely many solutions?

Expand: $3ax - 6 = 12x - 6$. The constants already agree, so the x -coefficients must agree: $3a = 12$, so $a = 4$.

Example 2 — Solve for the expression, not the variable

If $4a + 3b = 18$ and $a + 2b = 7$, what is the value of $a + b$?

Add the equations: $5a + 5b = 25$, so $a + b = 5$. (Check: $a = 3$, $b = 2$.)

Example 3 — Words → inequality → whole number

A rental costs a \$35 fee plus \$12 per hour. Maria can spend at most \$100. What is the greatest whole number of hours she can rent?

$35 + 12h \leq 100 \Rightarrow h \leq \frac{65}{12} \approx 5.42$. Hours must be whole, so round down: $h = 5$ (cost \$95; 6 hours would cost \$107).

6 · PRACTICE SET — 30 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1.

EASY · NO CALCULATOR NEEDED

If $6x - 15 = 33$, what is the value of $2x - 5$?

(A) 8

(B) 11

(C) 16

(D) 33

2.

EASY · NO CALCULATOR NEEDED

The formula $C = \frac{5}{9}(F - 32)$ converts a temperature of F degrees Fahrenheit to C degrees Celsius. Which of the following gives F in terms of C ?

(A) $F = \frac{5}{9}C + 32$ (B) $F = \frac{9}{5}C - 32$ (C) $F = \frac{9}{5}C + 32$ (D) $F = \frac{9}{5}(C - 32)$ **3.**

EASY · CALCULATOR

A mobile data plan costs a one-time activation fee of \$24 plus \$7.50 for each gigabyte (GB) of data used. The total cost for one customer was \$69. How many GB of data did the customer use?

4.

EASY · NO CALCULATOR NEEDED

Which of the following values of x satisfies the inequality $3 - 2x > 11$?

(A) -6

(B) -4

(C) 0

(D) 4

5.

EASY • NO CALCULATOR NEEDED

How many solutions does the equation $3(2x + 4) = 6x + 7$ have?

- (A) Exactly two solutions
- (B) Infinitely many solutions
- (C) Exactly one solution
- (D) No solution

6.

MEDIUM • NO CALCULATOR NEEDED

The sum of four consecutive odd integers is 112. What is the greatest of the four integers?

- (A) 27 (B) 29
(C) 31 (D) 25

7.

MEDIUM • NO CALCULATOR NEEDED

What is the solution to $\frac{2x-1}{3} - \frac{x+2}{4} = 1$?

8.

MEDIUM · NO CALCULATOR NEEDED

$$5x + 3y = 29$$

$$3x + 5y = 19$$

The solution to the system of equations above is (x, y) . What is the value of $x + y$?

(A) 3

(B) 6

(C) 24

(D) 48

9.

MEDIUM · NO CALCULATOR NEEDED

The sum S of the first n terms of an arithmetic sequence with first term a and last term ℓ is given by $S = \frac{n}{2}(a + \ell)$. Which of the following expresses a in terms of S , n and ℓ ?

(A) $a = \frac{2S - \ell}{n}$

(B) $a = \frac{S}{2n} - \ell$

(C) $a = \frac{2S}{n} + \ell$

(D) $a = \frac{2S}{n} - \ell$

10.

MEDIUM · NO CALCULATOR NEEDED

$$(k - 3)x + 8 = 2x + 3k - 1$$

In the equation above, k is a constant. For what value of k does the equation have no solution?

11.

MEDIUM · NO CALCULATOR NEEDED

For a fundraiser, the number of tickets t sold must be at least 15 less than three times the number of volunteers v . Which inequality represents this requirement?

(A) $t \geq 3v - 15$

(B) $t \leq 3v - 15$

(C) $t \geq 3(v - 15)$

(D) $t \geq 15 - 3v$

12.

MEDIUM · CALCULATOR

The table shows three values of the linear function g . For what value of x is $g(x) = -10$?

x	3	7	11
$g(x)$	14	30	46

(A) -2

(B) -3

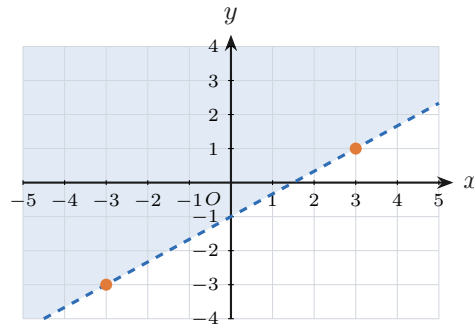
(C) $-\frac{5}{2}$

(D) 3

13.

MEDIUM · DESMOS

The shaded region in the xy -plane shows the solution set of an inequality. The boundary line is dashed. Which of the following inequalities defines the shaded region?



(A) $y \geq \frac{2}{3}x - 1$
 (C) $y > \frac{2}{3}x - 1$

(B) $y > \frac{3}{2}x - 1$
 (D) $y < \frac{2}{3}x - 1$

14.

HARD · NO CALCULATOR NEEDED

$$4(ax + 3) - 2(x - b) = 18x + 20$$

In the equation above, a and b are constants. If the equation is true for all values of x , what is the value of ab ?

15.

HARD · NO CALCULATOR NEEDED

$$3(kx - 2) = 12x + m$$

In the equation above, k and m are constants. Consider the following statements.

I. If $k = 4$ and $m = -6$, the equation has infinitely many solutions.

II. If $k \neq 4$, the equation has exactly one solution for every value of m .

III. If $m \neq -6$, the equation has no solution for every value of k .

Which of the statements must be true?

(A) I and II only

(B) I, II, and III

(C) I only

(D) II only

16.

HARD · NO CALCULATOR NEEDED

If $\frac{5}{3}(6w - 9) + 2 = 27$, what is the value of $2w - 3$?

(A) 8

(B) 25

(C) 4

(D) 5

17.

HARD · NO CALCULATOR NEEDED

$$3(p - 2q) = 4(r - p)$$

Which of the following expresses p in terms of q and r ?

(A) $p = \frac{6q - 4r}{7}$

(B) $p = \frac{4r + 6q}{5}$

(C) $p = \frac{4r + 6q}{7}$

(D) $p = \frac{4r - 6q}{7}$

18.

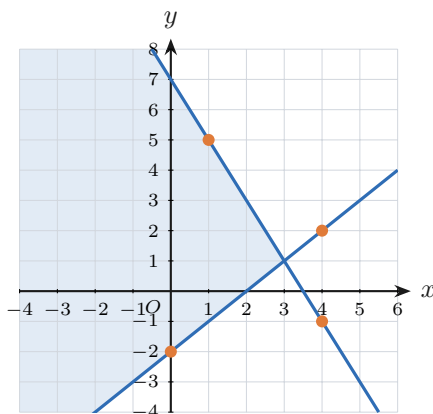
HARD · NO CALCULATOR NEEDED

What is the sum of all integers x that satisfy $-5 \leq \frac{4 - 3x}{2} < 8$?

19.

HARD · DESMOS

The shaded region in the xy -plane is the solution set of a system of two linear inequalities, and both boundary lines are solid. Which of the following systems has this solution set?



(A)
$$\begin{cases} y \geq -2x + 7 \\ y \leq x - 2 \end{cases}$$

(B)
$$\begin{cases} y \geq x - 2 \\ y \leq -2x + 7 \end{cases}$$

(C)
$$\begin{cases} y \leq x - 2 \\ y \leq -2x + 7 \end{cases}$$

(D)
$$\begin{cases} y \geq x - 2 \\ y \geq -2x + 7 \end{cases}$$

20.

HARD · CALCULATOR

A club sells T-shirts for \$18 each, and each T-shirt costs the club \$7 to make. It sells mugs for \$12 each, and each mug costs the club \$5 to make. The club also pays a fixed booth fee of \$150. The club sells 27 T-shirts. What is the least whole number of mugs it must sell to make a profit of at least \$800?

21.

HARD · NO CALCULATOR NEEDED

If $1 < x < 6$ and $-2 < y < 3$, which of the following could *not* be the value of $2x - 3y$?

(A) -6.5 (B) 0 (C) 17.5 (D) -7 **22.**

HARD · NO CALCULATOR NEEDED

Four consecutive multiples of 3 are listed in increasing order. Three times the sum of the two smallest of them equals the largest increased by 60. What is the largest of the four numbers?

23.

HARD · NO CALCULATOR NEEDED

The linear functions f and g are defined by the tables below. For one value of t , the equation $f(x) = g(x)$ has no solution. What is that value of t ?

x	0	2
$f(x)$	5	11

x	1	3
$g(x)$	4	t

(A) 10

(B) 13

(C) 16

(D) 7

24.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$c(2x - 5) + d = 8x + cd$$

In the equation above, c and d are constants. If the equation is true for all values of x , what is the value of d ?

25.

SAT CHALLENGE · CALCULATOR

$$(k + 2)x = 36$$

In the equation above, k is an integer. For how many values of k is the solution x a positive integer less than 10?

26.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$a(x - 4) \geq 6 - 2x$$

In the inequality above, a is a constant. The solution set of the inequality is all real numbers x such that $x \leq \frac{9}{2}$. What is the value of a ?

27.

SAT CHALLENGE · DESMOS

In the xy -plane, how many points (x, y) with positive integer coordinates satisfy both $3x + 2y \leq 17$ and $x - y \geq -2$?

28.

SAT CHALLENGE · CALCULATOR

A tank holds 120 liters of a salt solution that is 30% salt by volume. Some liters of the solution are drained from the tank and replaced with the same number of liters of a solution that is 80% salt, so that the tank again holds 120 liters. The resulting solution is 50% salt. How many liters were drained?

29.

SAT CHALLENGE · CALCULATOR

For each integer t , let P_t be the point $(t + 1, 2t - 3)$ in the xy -plane. For how many integer values of t does P_t lie in the solution set of the system $3x - 4y > 10$ and $x + 3y \geq -22$?

(A) 2 (B) 3

(C) 4 (D) 5

30.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$k^2x - 4k = 9x - 12$$

In the equation above, k is a constant. Which of the following statements is true?

(A) $k = 3$: infinitely many; others: one (B) No solution for both $k = 3$ and $k = -3$ (C) $k = 3$: infinitely many; $k = -3$: none (D) $k = -3$: infinitely many; $k = 3$: none